

**SUBSTITUTE FOR
SENATE BILL NO. 505**

A bill to amend 1994 PA 451, entitled
"Natural resources and environmental protection act,"
(MCL 324.101 to 324.90106) by adding part 151.

THE PEOPLE OF THE STATE OF MICHIGAN ENACT:

PART 151

STATEWIDE MICROPLASTICS RESEARCH AND MONITORING PLAN

**Sec. 15101. (1) The department, in consultation with other
state departments and agencies and interested entities, shall
develop a statewide microplastics research and monitoring plan to
assess the occurrence of microplastics, including in plants and
animals, and the impact of microplastics on human and animal
health, the health of the Great Lakes, and the critical drinking
water supply and identify proposed solutions to address the impacts**



1 of microplastics, to the extent feasible.

2 (2) The department may enter into 1 or more contracts with
3 research institutions in this state, including those associated
4 with Grand Valley State University, Michigan State University, the
5 University of Michigan, Wayne State University, and other academic
6 and research institutions that have demonstrated expertise relating
7 to the effects of microplastics on the Great Lakes, for research
8 services that will contribute to the statewide microplastics
9 research and monitoring plan.

10 (3) The statewide microplastics research and monitoring plan
11 must include, but is not limited to, all the following:

12 (a) Strategies for developing a comprehensive plan to identify
13 and evaluate the ecological, health, and economic impacts
14 associated with microplastics in the Great Lakes.

15 (b) Methods for sampling, detecting, and characterizing
16 microplastics that are consistent with recognized standards from
17 scientific, peer-reviewed research or relevant agencies, including,
18 but not limited to, any of the following:

19 (i) The International Joint Commission.

20 (ii) The United States Environmental Protection Agency.

21 (iii) The National Oceanic and Atmospheric Administration.

22 (iv) ASTM International.

23 (c) A list of the baseline concentrations of microplastics in
24 the Great Lakes that includes, but is not limited to, all the
25 following information:

26 (i) An assessment of the associated environmental and human
27 health impacts.

28 (ii) Solutions this state can immediately implement.

29 (iii) Factors that should be considered, such as microplastic



1 particle age, size, shape, type, and location.

2 (d) Research actions that are in alignment with best practices
3 and actions taken by other Great Lakes states and Canada.

4 (e) An investigation into the sources of microplastics and the
5 significance of pathways related to environmental and human health
6 impacts of microplastics.

7 (f) A list of long-term research objectives and strategies for
8 reducing microplastics in the Great Lakes, which must be updated at
9 least once every 5 years and made available to the public.

10 (g) An impact assessment framework for microplastics
11 consistent with best available science and research, and based on
12 the best available information on exposure of microplastics to
13 organisms, including humans, through pathways identified under
14 subdivision (e).

15 (h) Approaches to reduce the introduction of microplastics
16 into the Great Lakes from significant pathways and sources, with an
17 emphasis on the sizes, shapes, and types of microplastics that are
18 associated with significant environmental and human health impacts.

19 (i) Recommendations for both of the following:

20 (i) Policy changes, including statutory changes, or additional
21 research that may be needed.

22 (ii) Solutions to control microplastic introduction into the
23 Great Lakes, such as source elimination, pathway intervention, and
24 public education and outreach.

25 (j) A timeline to implement the statewide microplastics
26 research and monitoring plan and reduce the impacts of
27 microplastics on the Great Lakes.

28 (k) A list of potential barriers to implementation that
29 includes estimated costs and potential funding sources.



1 (4) To develop and implement the statewide microplastics
2 research and monitoring plan, the department may utilize existing
3 information and leverage ongoing federal efforts and efforts of
4 this state and other states, when possible.

5 (5) Not later than July 1, 2027, the department shall submit
6 the statewide microplastics research and monitoring plan to the
7 legislature. Not later than January 30, 2029, the department shall
8 submit a report to the legislature regarding the implementation of
9 the statewide microplastics research and monitoring plan.

10 (6) To the extent that funds are available from bonds and
11 other sources, including federal funds, not later than July 1,
12 2028, the department shall adopt and implement the statewide
13 microplastics research and monitoring plan.

14 (7) As used in this section:

15 (a) "Department" means the department of environment, Great
16 Lakes, and energy.

17 (b) "International Joint Commission" means a binational
18 organization established by the governments of the United States
19 and Canada under the Boundary Waters Treaty of 1909, and whose
20 responsibilities were expanded under the Great Lakes Water Quality
21 Agreement of 1972.

22 (c) "Microplastics" means solid polymeric materials that may
23 contain chemical additives or other substances and that are
24 particles less than 5,000 micrometers, in all 3 dimensions.
25 Microplastics do not include polymers that are derived in nature
26 and that have not been chemically modified, except by hydrolysis.